Course Name: Introduction to Computer and C Programming
Course Code: CS-111
Course Type: Institute Core

Contact Hours/Week: 2L/2P **Course Credits:** 03

Course Objectives

- To introduce the concept of computer fundamentals and computer programming.
- To enable the student to design algorithms.
- The enable the students to understand the "C" language and its application in problem-solving.

Unit Number	Course Content	Contact Hours
UNIT-01	**Programming Fundamentals**: Introduction to computer, block diagram and organization of computer, number system and binary arithmetic, processing data, hardware, software, firmware,	05L
UNIT-02	**Types of programming language**: Machine language, Assembly level language, higher level language, source file, object file, translators-assembler, compiler, interpreter. Evolution and classification of programming languages.	05L
UNIT-03	**Programming Techniques**: Steps in program development, algorithm, flowchart, pseudo-code.	06L
UNIT-04	**C Language**: 'C' character set, literals, keywords, identifiers, data types and size, variable declaration, expression, labels, statements, formatted input output statements, types of operators, data type conversion, mixed mode arithmetic, control structures.	08L
UNIT-05	**Control Statement and Derived Datatypes:**Control Statement and Expressions, Looping, Arrays and String, Pointers, Structures, Union.	05L
UNIT-06	**Functions and File Handling**: 'C' functions, library functions, parameter passing, recursion, 'C' files, function for file handling, 'C' pre-processors and command line arguments, macros and conditional compiler directives.	08L

Course Outcomes

Upon successful completion of the course, the students will be able to
CO1: Know the basic components of the computer and the working of each device.
CO2: Design algorithms and flowcharts
CO3: Understand the fundamentals of C programming.
CO4: Use suitable data structure for problem-solving

Text Books:

1. Briain W. Kenigham and Dennis Ritchie,C Programming Language, Prentice Hall of India.
2. Byron Gottfried,Programming with C, Tata McGraw Hill.
3. Herbert Schildt,The Complete Reference C, Tata McGraw Hill.

Reference Books:

1. Yashwant Kanetkar, Let us C, BPB Publication.
2. A. Forouzan and R.F. Gilberg, A Structured Programming Approach in C, Cengage Learning.
3. S.K.Srivastava and Deepali Srivastava,C In Depth, BPB Publications.